

Lab: 1

30.7.25

Exploring the Deep Learning Platforms

Various deep learning platforms.

Google Colab:

Free online Jupyter notebook with GPU/TPU

Jupyter notebook:

not a framework, but an interactive coding environment

combines codes, markdown, outputs

Difference between Google Colab & Jupyter

Google Colab	Jupyter Notebook
+ Cloud based platform	Local / Browser-based
+ Built in free GPU/TPU	GPU manual setup
+ Runs in browser	Anacoda + Jupyter
+ easy sharing	manual file sharing
+ Saves to Google	Saves to local system

Key Frameworks of PyTorch, TensorFlow:

TensorFlow:

Creator / Organisation: Google (2015)

main features: Scalable across CPUs, GPUs,
high-performance model training
integrated Keras API for simplicity

popular use cases: computer vision
NLP
ML

PyTorch

Organisation: Facebook AI Research (2016)

main features: dynamic computational graphs
native pythonic syntax
Strong GPU acceleration support

Popular uses: Research & Academic projects
NLP models
Fast model prototyping

Graph Type: dynamic

Conclusion:

Exploring different DL platforms helped to
understand their features, setup & usage.

~~13/11/25~~

Result:

These are the various platforms helped

lab1.ipynb (6) - JupyterLab x Breast Cancer Wisconsin (Diagn...) x Why use load_digits x Inbox (2,282) - th0591@srmist... x +

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Code Notebook Python 3 (ipykernel)

```
py-config are installed in '/home/jupyter-ra2311047010010/.local/bin' which is not on PATH.
Consider adding this directory to PATH or, if you prefer to suppress this warning, use --n
o-warn-script-location.
----- 11/26 [numpy] WARNING: The script markdown_py i
s installed in '/home/jupyter-ra2311047010010/.local/bin' which is not on PATH.
Consider adding this directory to PATH or, if you prefer to suppress this warning, use --n
o-warn-script-location.
----- 19/26 [tensorboard] WARNING: The script tensorb
oard is installed in '/home/jupyter-ra2311047010010/.local/bin' which is not on PATH.
Consider adding this directory to PATH or, if you prefer to suppress this warning, use --n
o-warn-script-location.
----- 20/26 [ml-dtypes] WARNING: The script markdown-
it is installed in '/home/jupyter-ra2311047010010/.local/bin' which is not on PATH.
Consider adding this directory to PATH or, if you prefer to suppress this warning, use --n
o-warn-script-location.
----- 25/26 [tensorflow] WARNING: The scripts import_
pb_to_tensorboard, saved_model_cli, tensorboard, tf_upgrade_v2, tf_lite_convert and toco are
installed in '/home/jupyter-ra2311047010010/.local/bin' which is not on PATH.
Consider adding this directory to PATH or, if you prefer to suppress this warning, use --n
o-warn-script-location.
----- 26/26 [tensorflow]
Successfully installed absl-py-2.3.1 astunparse-1.6.3 flatbuffers-25.2.10 gast-0.6.0 google-
pasta-0.2.0 grpcio-1.74.0 h5py-3.14.0 keras-3.11.1 libclang-18.1.1 markdown-3.8.2 markdown-i
t-py-3.0.0 mdurl-0.1.2 ml-dtypes-0.5.3 namex-0.1.0 numpy-2.1.3 opt-einsum-3.4.0 optree-0.17.
0 protobuf-5.29.5 rich-14.1.0 tensorboard-2.19.0 tensorboard-data-server-0.7.2 tensorflow-2.
19.0 tensorflow-io-gcs-fsfilesystem-0.37.1 termcolor-3.1.0 werkzeug-3.1.3 wrapt-1.17.2
Note: you may need to restart the kernel to use updated packages.

[2]: import tensorflow as tf

2025-08-07 21:01:43.474084: I external/local_xla/xla/tsl/cuda/cudart_stub.cc:32] Could not f
```

VARIABLES
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Simple 1 s 5 Python 3 (ipykernel) | Idle Mem: 1.02 GB Mode: Command Ln 5, Col 9 lab1.ipynb 1

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Code Notebook Python 3 (ipykernel)

```
[1]: pip install tensorflow
```

Defaulting to user installation because normal site-packages is not writeable
Collecting tensorflow
 Downloading tensorflow-2.19.0-cp310-cp310-manylinux_2_17_x86_64.manylinux2014_x86_64.whl.m
 etadata (4.1 kB)
Collecting absl-py<=1.0.0 (from tensorflow)
 Downloading absl_py-2.3.1-py3-none-any.whl.metadata (3.3 kB)
Collecting astunparse>=1.6.0 (from tensorflow)
 Downloading astunparse-1.6.3-py2.py3-none-any.whl.metadata (4.4 kB)
Collecting flatbuffers>=24.3.25 (from tensorflow)
 Downloading flatbuffers-25.2.10-py2.py3-none-any.whl.metadata (875 bytes)
Collecting gast!=0.5.0,!0.5.1,!0.5.2,>=0.2.1 (from tensorflow)
 Downloading gast-0.6.0-py3-none-any.whl.metadata (1.3 kB)
Collecting google-pasta>=0.1.1 (from tensorflow)
 Downloading google_pasta-0.2.0-py3-none-any.whl.metadata (814 bytes)
Collecting libclang>=13.0.0 (from tensorflow)
 Downloading libclang-18.1.1-py2.py3-none-manylinux2010_x86_64.whl.metadata (5.2 kB)
Collecting opt_einsum>=2.3.2 (from tensorflow)
 Downloading opt_einsum-3.4.0-py3-none-any.whl.metadata (6.3 kB)
Requirement already satisfied: packaging in /opt/tljh/user/lib/python3.10/site-packages (fro
m tensorflow) (24.0)
Collecting protobuf!=4.21.0,!4.21.1,!4.21.2,!4.21.3,!4.21.4,!4.21.5,<6.0.0dev,>=3.20.3
(from tensorflow)
 Downloading protobuf-5.29.5-cp38-abi3-manylinux2014_x86_64.whl.metadata (592 bytes)
Requirement already satisfied: requests<3,>=2.21.0 in /opt/tljh/user/lib/python3.10/site-pac
kages (from tensorflow) (2.31.0)
Requirement already satisfied: setuptools in /opt/tljh/user/lib/python3.10/site-packages (fr
om tensorflow) (65.6.3)
Requirement already satisfied: six>=1.12.0 in /opt/tljh/user/lib/python3.10/site-packages (f

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Notebook Python 3 (ipykernel)

W0000 00:00:1754580703.555856 121210 computation_placer.cc:177] computation placer already registered. Please check linkage and avoid linking the same target more than once.
W0000 00:00:1754580703.555877 121210 computation_placer.cc:177] computation placer already registered. Please check linkage and avoid linking the same target more than once.
W0000 00:00:1754580703.555880 121210 computation_placer.cc:177] computation placer already registered. Please check linkage and avoid linking the same target more than once.
W0000 00:00:1754580703.555883 121210 computation_placer.cc:177] computation placer already registered. Please check linkage and avoid linking the same target more than once.
2025-08-07 21:01:43.563589: I tensorflow/core/platform/cpu_feature_guard.cc:210] This TensorFlow binary is optimized to use available CPU instructions in performance-critical operations.
To enable the following instructions: AVX2 FMA, in other operations, rebuild TensorFlow with the appropriate compiler flags.

[3]: `print(tf.__version__)`

2.19.0

[4]: `import tensorflow as tf
a = tf.constant([[1, 2], [3, 4]])
b = tf.constant([[5, 6], [7, 8]])
c = tf.add(a, b)
print(c)`

tf.Tensor(
[[6 8]
[10 12]], shape=(2, 2), dtype=int32)

2025-08-07 21:02:11.316893: E external/local_xla/xla/stream_executor/cuda/cuda_platform.cc:51] failed call to cuInit: INTERNAL: CUDA error: Failed call to cuInit: UNKNOWN ERROR (303)

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